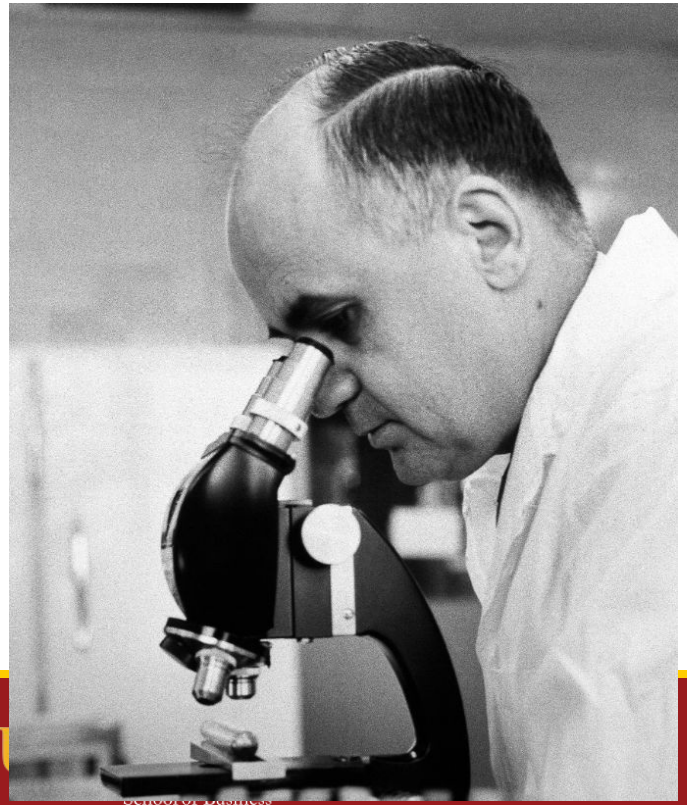
# Microeconomics for Business

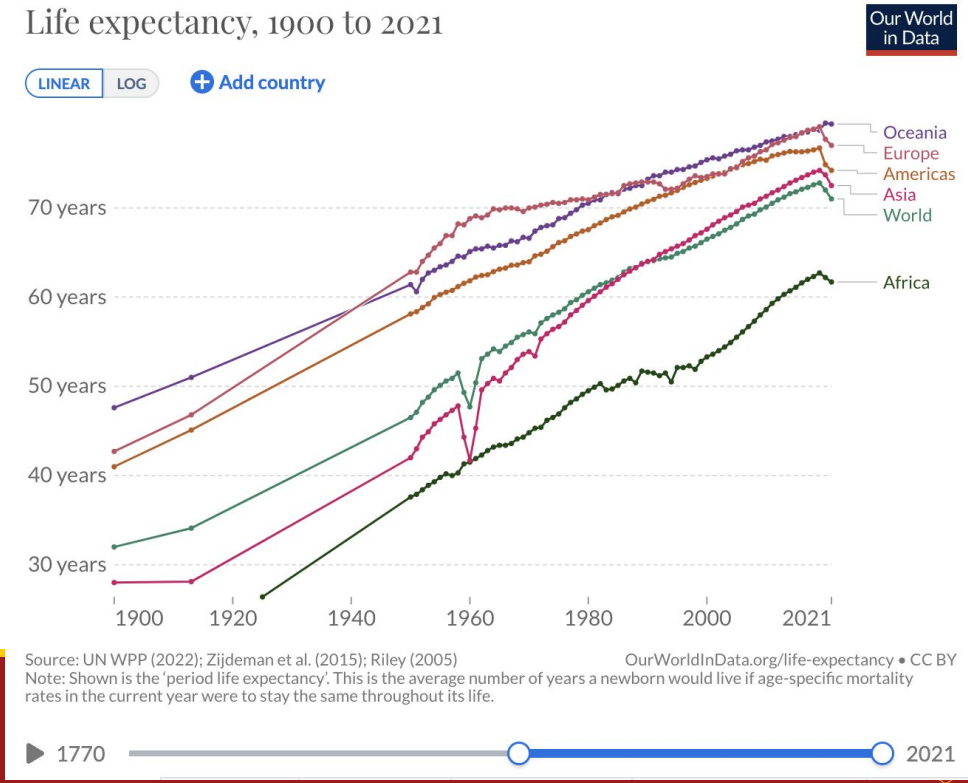
Chapter 3 – Production Choices

Firms
and
A Perfectly Competitive Market

# Person who had the biggest impact of the 20th Century?

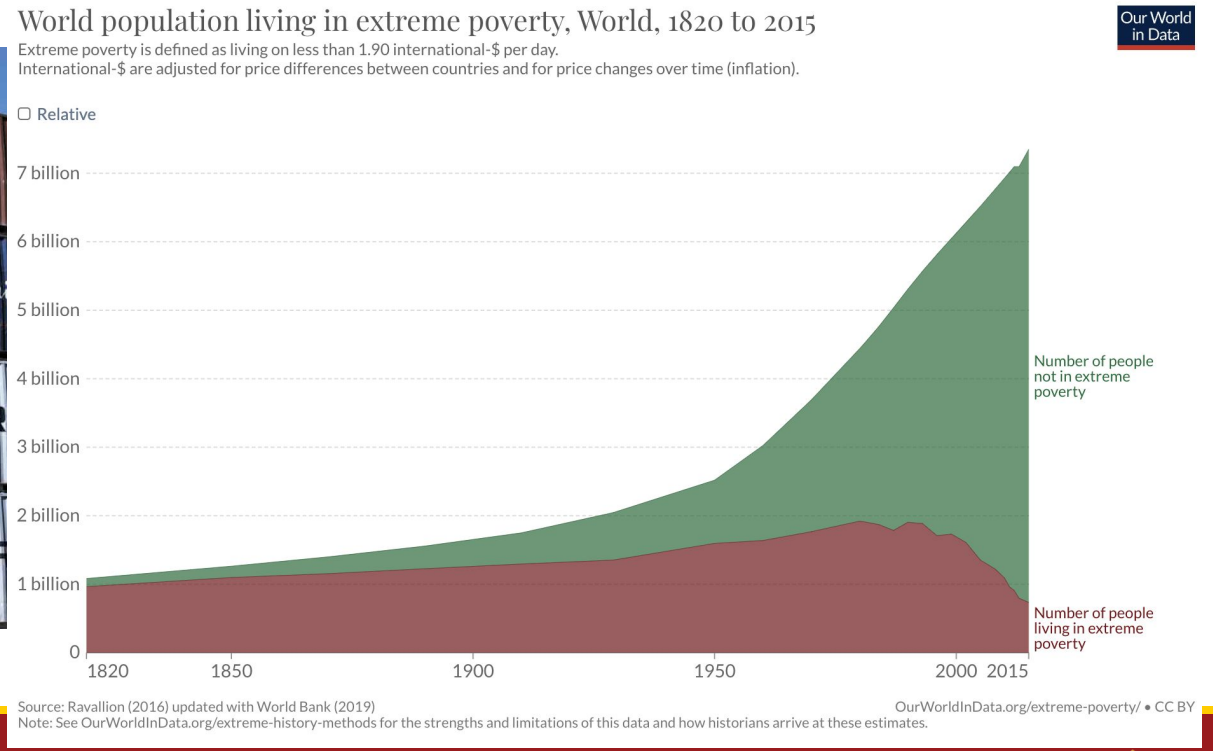
# Heroes of the 20th Century





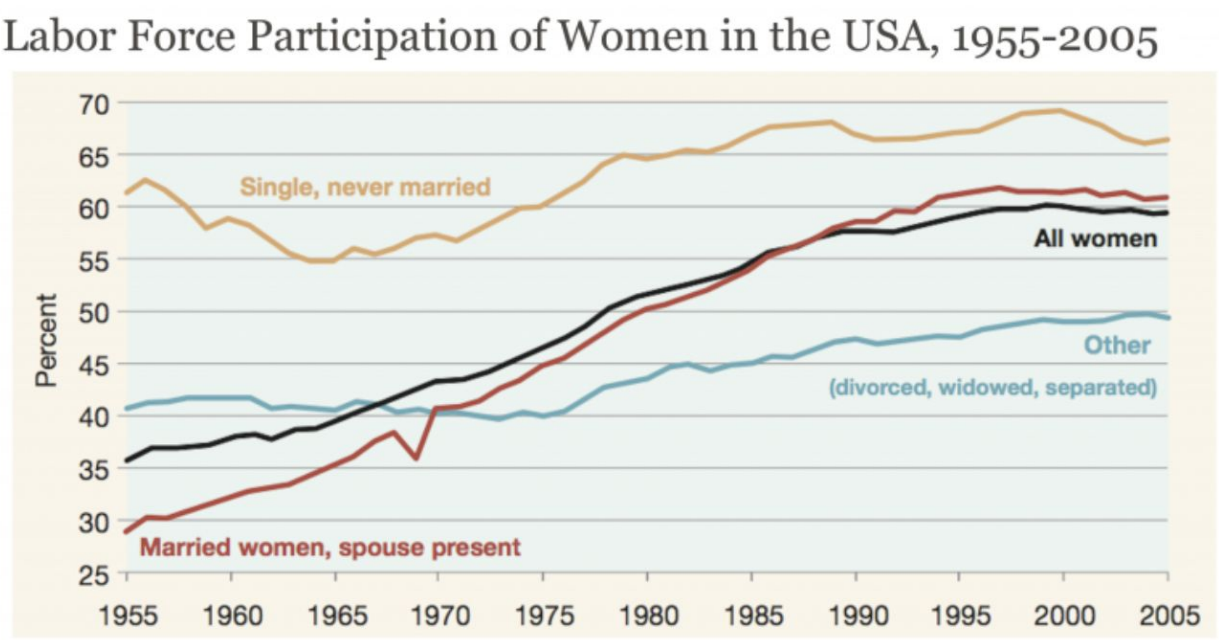
# Heroes of the 20th Century



# Heroes of the 20th Century



# Firms

Firm: business unit, company or enterprise.

Most production of goods/services done by firms

# Firms



# Firms

Think about how they operate?

Choices: how much of the different inputs to use?
how much output to produce?

Objective: to maximize profits

# Microeconomics for Business

Chapter 3 – Let's talk about costs

# My sister-in-law's coffee shop made 80,000 in profit last year, what is Keyana's economic profit? (hypothetical, I don't ask actually ask her how much she makes)



More than 80,000 0%

80,000 0%

Less than 80,000 0%

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# Accounting profit v. Economic profit

Economic cost takes into account the **opportunity cost**

"could have gained"

What is the cost of Keyana's time to run the coffee shop? What could she have done instead?

Office Space (Capital $K$)

## Office Space (Capital $K$)

### Scenario 1

Firm owns the office space

Does not need to pay rent

## Office Space (Capital $K$)

### Scenario 1

Firm owns the office space

Does not need to pay rent

### Scenario 2

Firm rents the office space from a landlord

rent $r = 2{,}000$

## Office Space (Capital $K$)

### Scenario 1

Firm owns the office space

Does not need to pay rent

There is an opportunity cost $r = 2{,}000$

### Scenario 2

Firm rents the office space from a landlord

rent $r = 2{,}000$

# Profitable business?

# Profitable business?

## Scenario 1

Firm owns the office space

Does not need to pay rent

# Profitable business?

## Scenario 1

Firm owns the office space

Does not need to pay rent

Revenue + $3,000

# Profitable business?

## Scenario 1

Firm owns the office space

Does not need to pay rent

| | |
|---|---|
| Revenue | $+ \$3{,}000$ |
| Wages | $- \$2{,}000$ |

# Profitable business?

## Scenario 1

Firm owns the office space

Does not need to pay rent

| | |
|---|---|
| Revenue | + $3,000 |
| Wages | − $2,000 |
| Actual Rent | $0 |
| Profit | + $1,000 |

# Profitable business?

## Scenario 1

Firm owns the office space

Does not need to pay rent

| | |
|---|---|
| Revenue | + $3,000 |
| Wages | − $2,000 |
| Op. Cost Rent | −$2,000 |
| Profit | − $1,000 |

# Economic versus Accounting Cost case study




In 2004, Jorgen Knudstorp took over as CEO of the Lego group.

The company had positive accounting profit from 1993 to 2002, but when Knudstorp included the cost of capital, the company had been running negative economic profit.

The backers of Lego would have been better off just putting their money in government bonds.

# Speaking of Lego—



ECONOMY

## It may be more lucrative to invest in collectible LEGO sets than in gold, study finds

December 21, 2021 · 5:29 PM ET

Heard on All Things Considered

By Jason Fuller, Courtney Dorning

## Investing in Lego more l gold, study suggests

**Market for secondhand Lego also rises at a better rate annually than stocks, bonds, stamps, art and wine**

## LEGOs are worth more than gold




$500 in 2007

$2400 in 2023

# Economic versus Accounting Cost

- Not every set in the Millenium Falcon, study found an average annual return of 11%

What is the opportunity cost of buying a lego set and holding onto it to sell for profit?

# Economic versus Accounting Cost

- Can’t invest the money elsewhere (an asset that rises with inflation would go from $100 to $120 from 2018 to 2023)
- Storage cost ($18 over five years)
- My time, buying the lego, reselling it, dusting the box ($50)

So is it worth it?

# Economic versus Accounting Cost

- Suppose I bought a 100 dollar set in 2018 at 11% annual increase, I'd sell it for $152
- $52 in profit vs. ($20 cost of capital + $18 cost of storage + $50 cost of time= $88)

# ECON LIFE HACK

Know the value of your time, so you don’t waste it.

# Problem

You have started a small business and have the following data from the first year of operation:

- Total revenue: $120,000
- Costs for materials and labor (explicit costs): $60,000
- You quit a job that was paying you $50,000 per year to start this business.
- You invested $10,000 that was previously earning 5% annual interest in a savings account.

Calculate both the accounting profit and the economic profit for the business in its first year.

# Types of Costs

Fixed

Variable

Marginal

Average

# Fixed Costs

Costs firm pays regardless of quantity produced.

Examples– Rent, Interest payments, Salary (to salaried workers)

In the long run there are no fixed costs (firm can change)

# Fixed, Variable and Total Cost in the Short-Run

Cost = Fixed Cost + Variable Cost

$$C(q) = FC + VC(q)$$

When poll is active, respond at **pollev.com/matthewhill691**

Text **MATTHEWHILL691** to **37607** once to join

# It's been raining and my sister-in-law's coffee shop is slow. She's thinking about closing early, but then she remembers that rent is $8000. Should she still close early?



Yes

No

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# Fixed and Sunk Costs

Fixed Costs are similar to sunk costs.

They should not enter into the production decision.

The money is already gone.

# Knowing when to quit is a valuable skill

"You'll hear people say, 'If I quit now, I'll have wasted my time.' That's a sunk-cost fallacy problem. The time that you've already spent is gone—what matters is if you're wasting more time by continuing with the project."

- Annie Duke

Make sure your identity is not wrapped up in a project/company/person. Makes quitting that thing difficult

# Sunk Cost Fallacy (Marginal Thinking)

- “Do not throw good money after bad”
- Shouldn’t make decisions based on costs you cannot recoup

# Variable Costs

Cost = Fixed Cost + Variable Cost

$$C(q) = FC + VC(q)$$

# Average and Marginal Cost

Average Cost $AC = \frac{C}{q}$

# Average and Marginal Cost

Average Cost $$AC = \frac{C}{q}$$

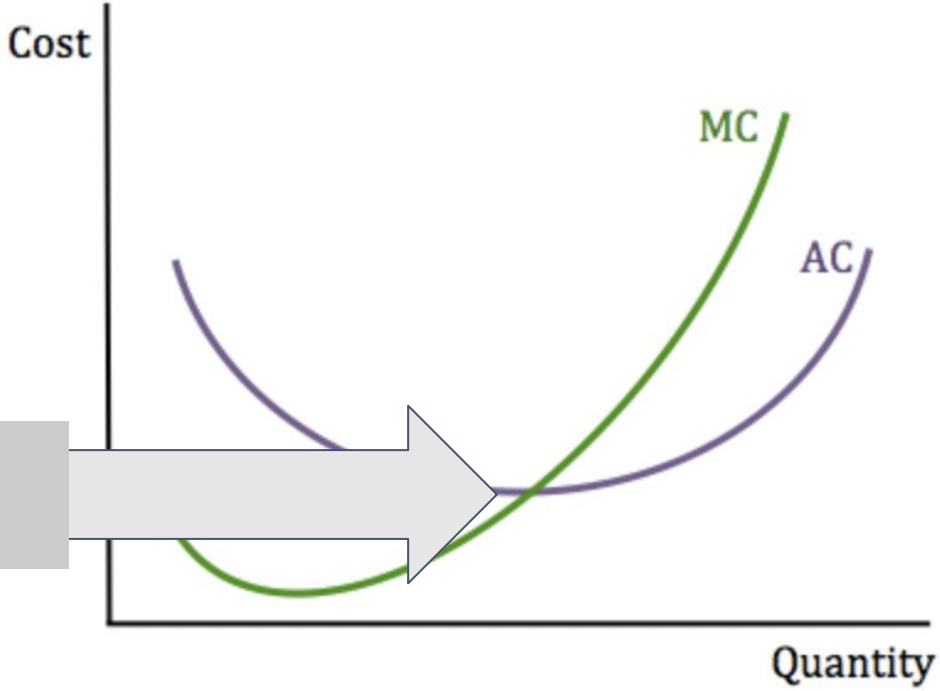
Marginal Cost $$MC = \frac{dC}{dq}$$

# Average and Marginal Cost

Example–

A player's scoring average is like the average cost. The amount the scored last game is like the marginal cost.

MC = ATC
At ATC minimum



# Cost Structures in the Real World





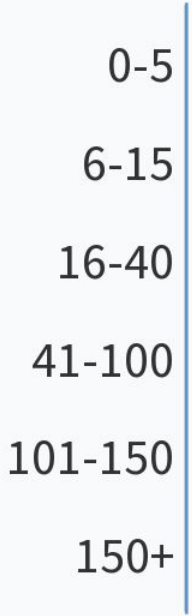
# Upward sloping MC curves

Why does marginal cost slope upwards?

When poll is active, respond at **pollev.com/matthewhill691**

Text **MATTHEWHILL691** to **37607** once to join

# How many people do you know?



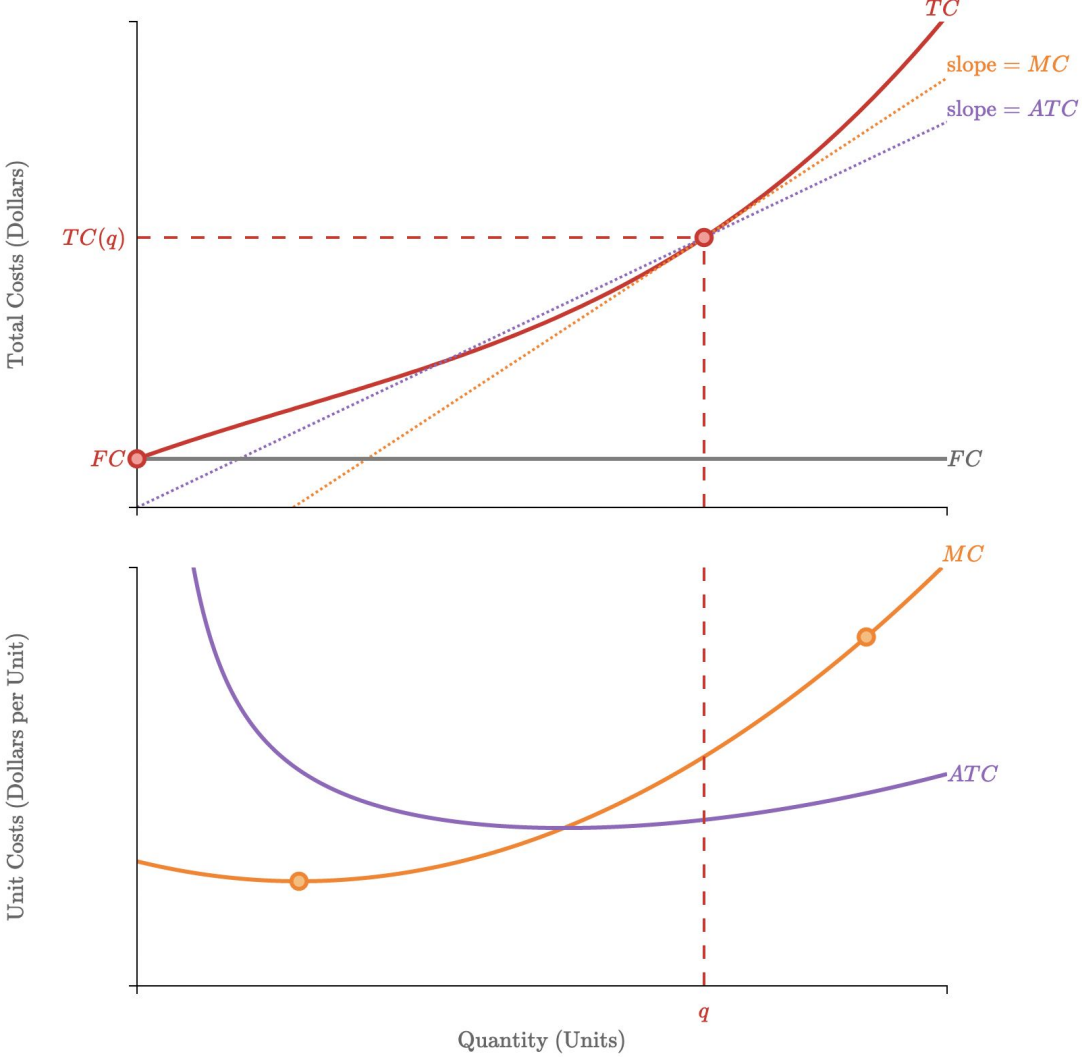

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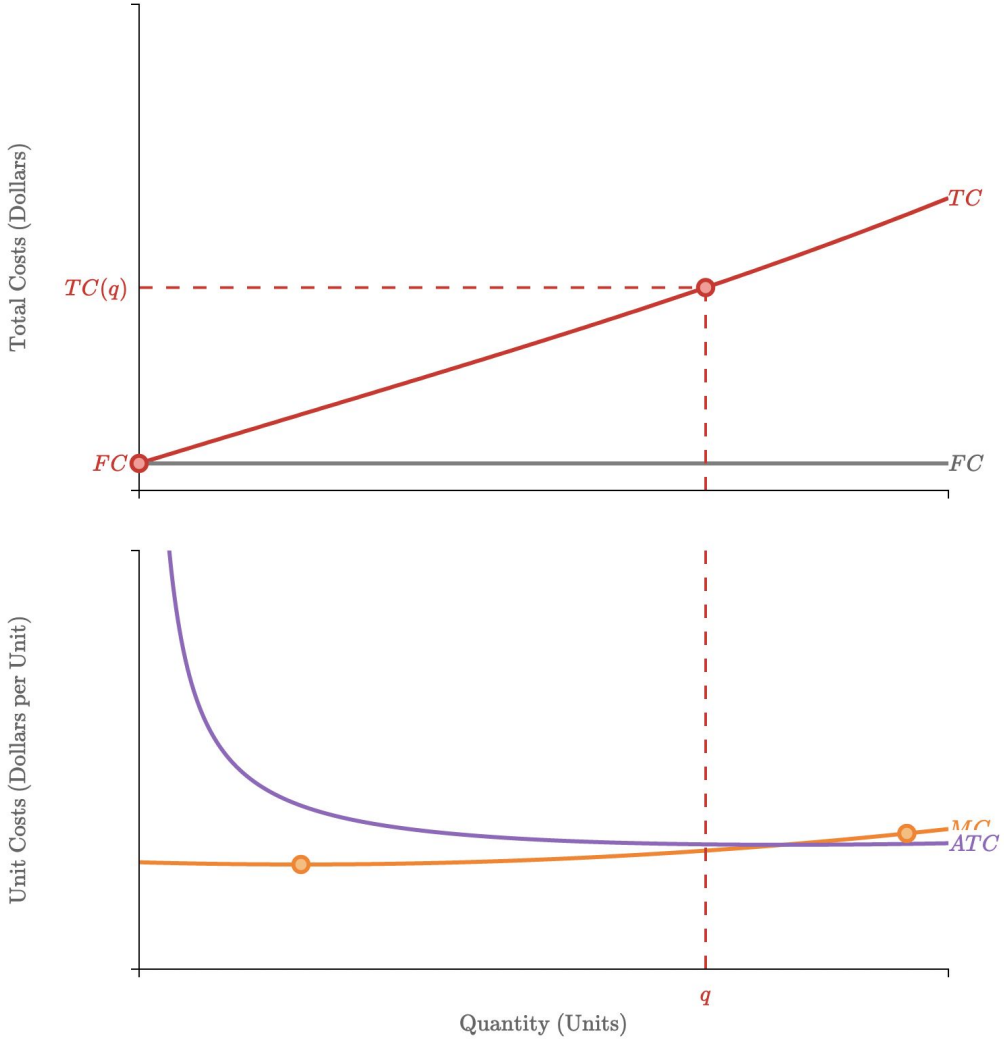
# Dunbar's Number

- Dunbar postulated that number of people one could maintain a social relationship with was 150 (basically you'd be comfortable grabbing coffee with)
- Many primate groups are below this size
- Organizations above this size require considerable bureaucracy to manage its members
- Explains why start-ups can move fast, while large companies are slow to change (see also Innovator's Dilemma)

# Physical Production (scarce resources)



# Physical Production (automated, cheap)



# Cost Structures in the Real World

https://www.econgraphs.org/graphs/v1/micro/producer_theory/cost_curves?textbook=varian

# The Short-Run Versus the Long-Run

Short-Run: A situation in which there is at least one fixed input and some fixed cost. An actual situation. Time frame: right now, next couple months

Long-Run: A situation in which all inputs and costs are variable. A planning situation with free mobility. Time frame: more of a theoretical statement about where a firm/industry might end up

# Economies and Diseconomies of Scale

Economies of Scale

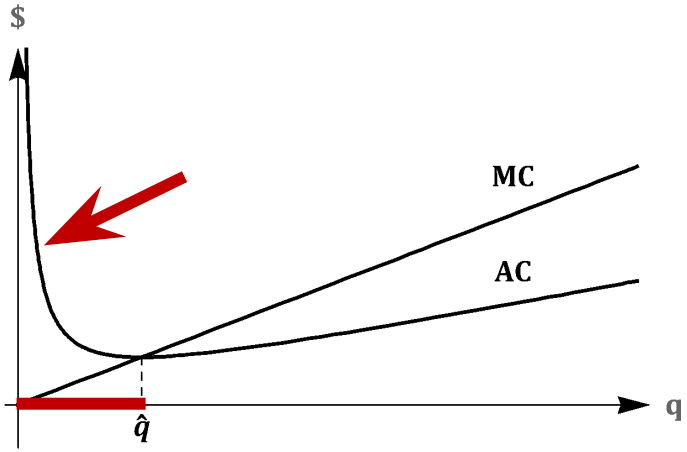
$$\frac{dAC(q)}{dq} < 0$$

Diseconomies of Scale

$$\frac{dAC(q)}{dq} > 0$$

Neither Economies nor Diseconomies of Scale

$$\frac{dAC(q)}{dq} = 0$$



# Economies and Diseconomies of Scale

Economies of Scale

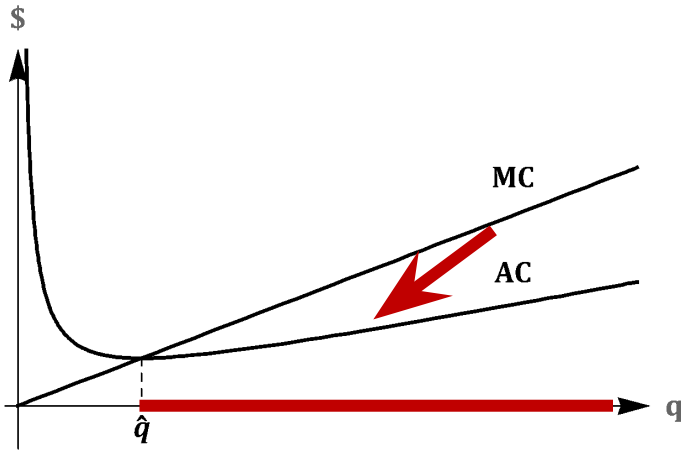
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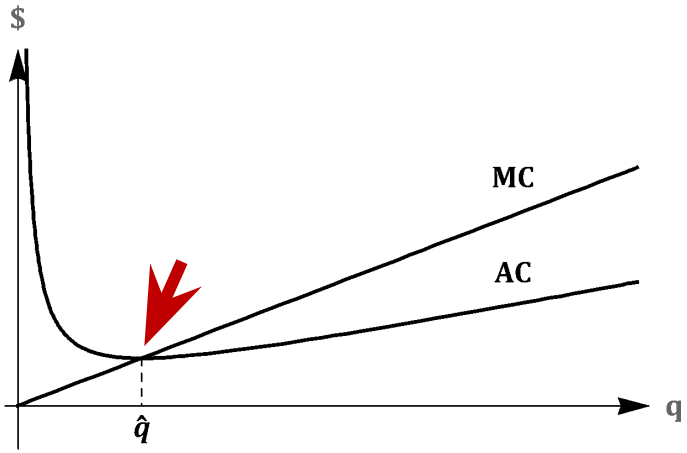


# Economies and Diseconomies of Scale

Economies of Scale $\frac{dAC(q)}{dq} < 0$

Diseconomies of Scale $\frac{dAC(q)}{dq} > 0$

Neither Economies nor Diseconomies of Scale $\frac{dAC(q)}{dq} = 0$



https://twitter.com/i/status/1695549490969665590

# The beauty of Economies of Scale

# Microeconomics for Business

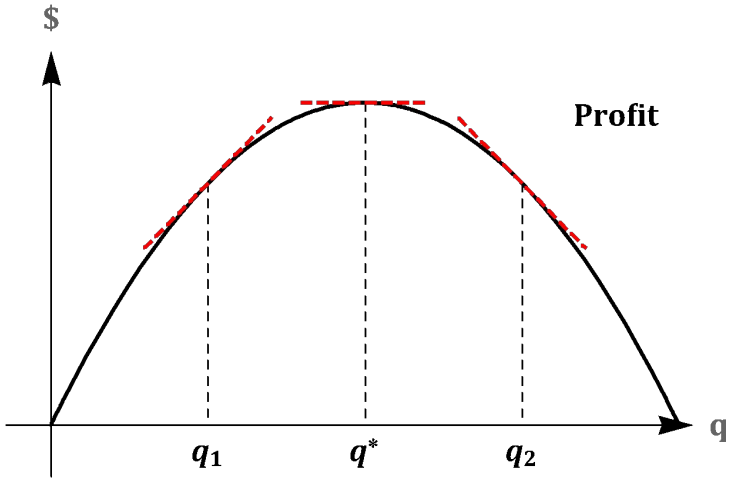
## Chapter 3 – Production Choices

## Maximizing Profit: The First-Order Condition

# What should the firm’s goal be?

First-order Condition:

Marginal Profit = 0



We can rewrite the condition:

Profit = Revenue - Cost

Marginal Profit = Marginal Revenue – Marginal Cost

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Profit = Revenue - Cost

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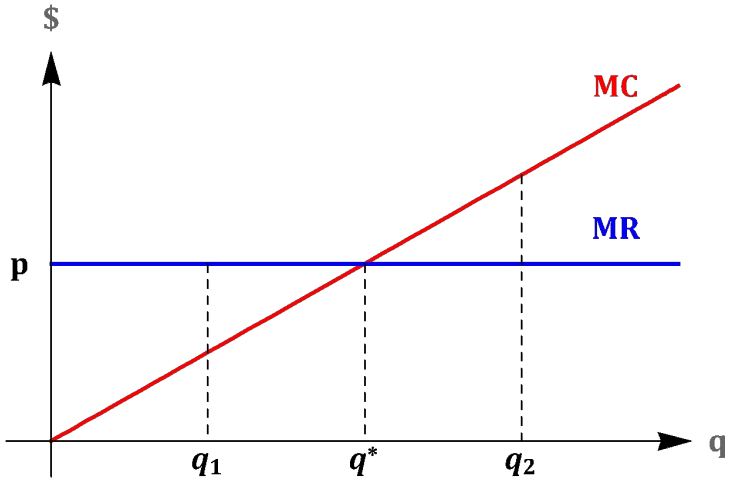
Marginal Profit = 0 → Marginal Revenue = Marginal Cost

# Under perfect Competition, MR is horizontal

$$Revenue = pq$$

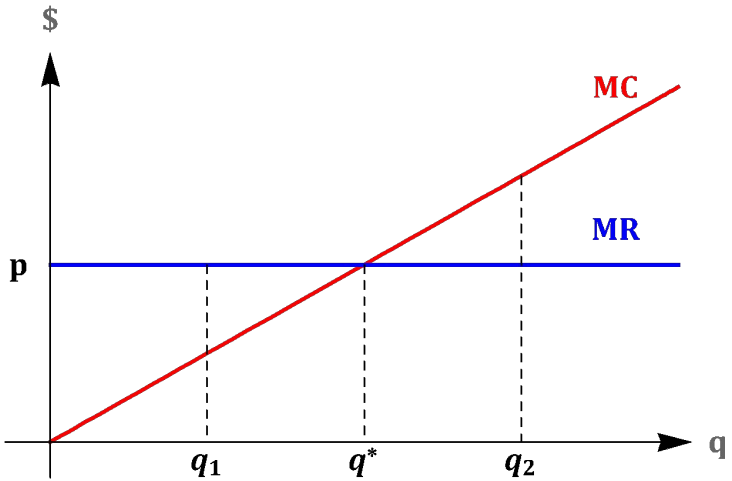
$$\frac{d\ Revenue}{dq} = p$$

$$MR = p$$



First-Order Condition

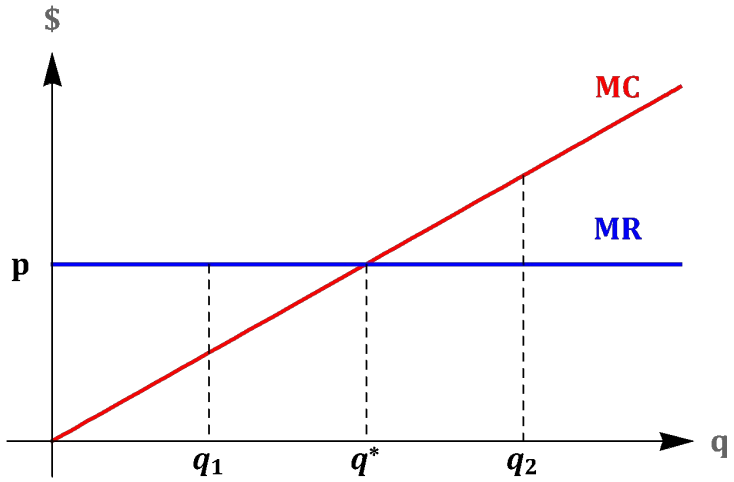
## Marginal Revenue = Marginal Cost



## First-Order Condition

MR= MC

p = MC



The First-Order Condition has nothing to do with the fixed cost

$$Profit(q) = Revenue - Cost$$



$$Profit(q) = pq - (FC + VC(q))$$

The First-Order Condition has nothing to do with the fixed cost, because *FC* does not vary with *q*.

$$Profit(q) = Revenue - Cost$$

$$Profit(q) = pq - (FC + VC(q))$$

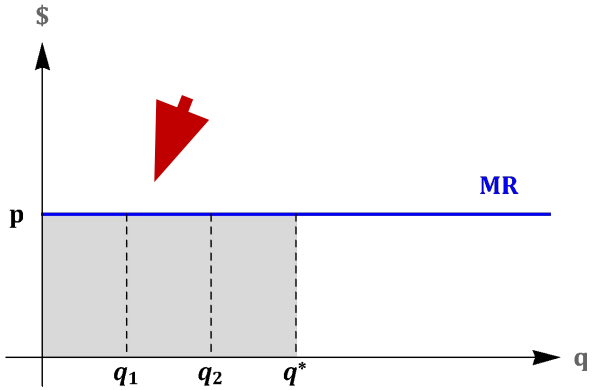
$$Profit(q) = pq - VC(q) - FC$$

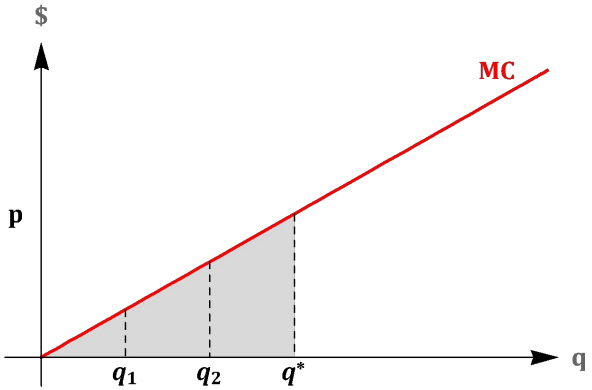
$$Profit(q) = Variable\ Profit(q) - FC$$

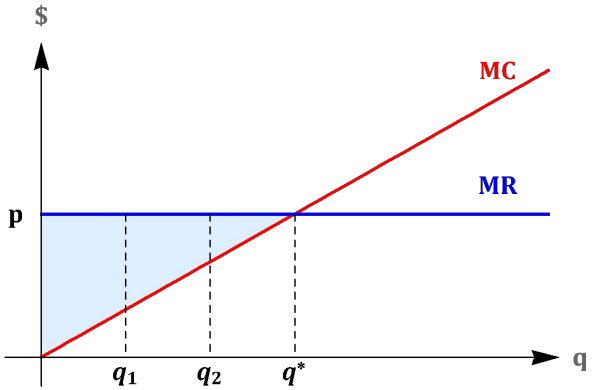
We will now present visuals of

1) Variable Profit
2) Total Profit

# 1) Use the MR and MC curves to illustrate variable profit

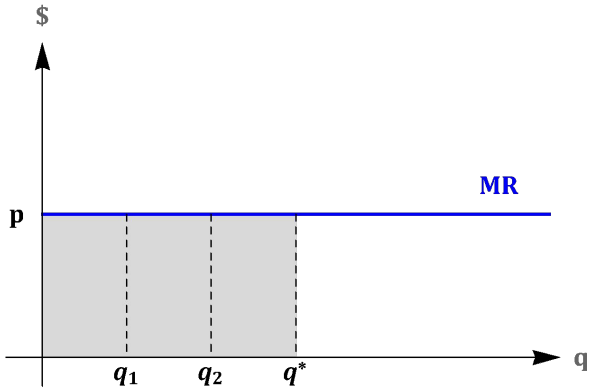


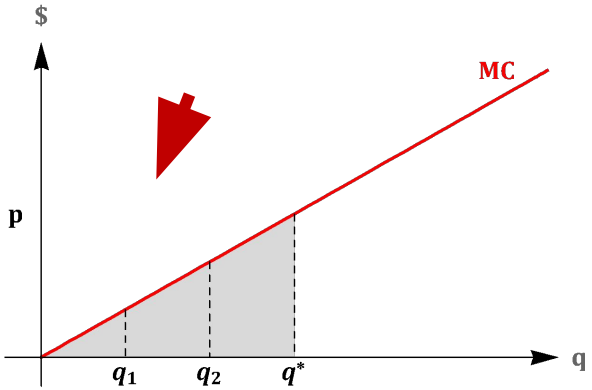



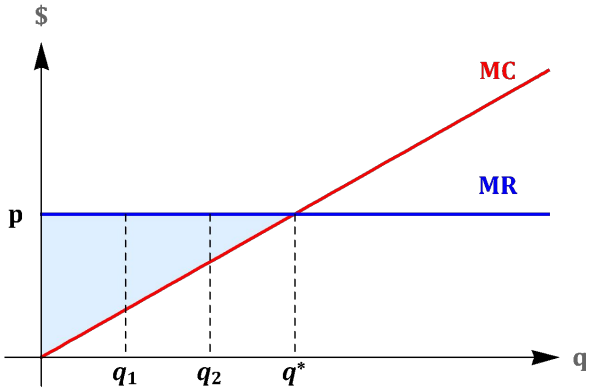




Revenue at $q^*$ $-$ Variable Cost at $q^*$ $=$ Variable Profit at $q^*$

# 1) Use the MR and MC curves to illustrate variable profit



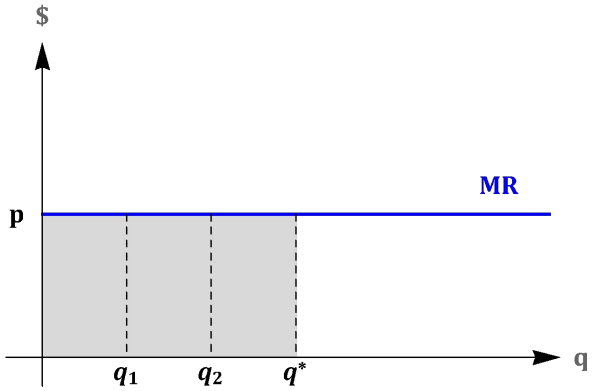



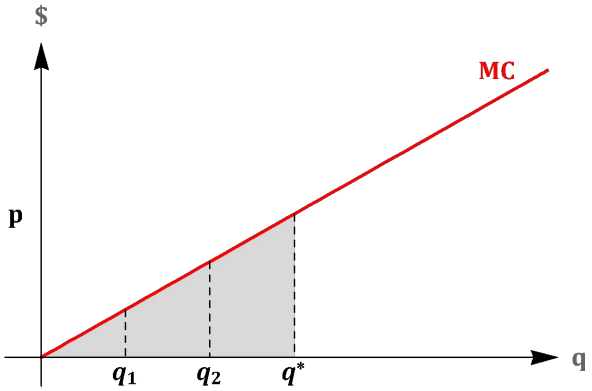



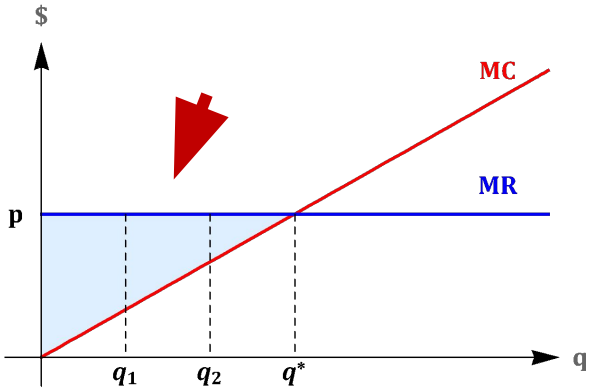

Revenue at $q^*$ − Variable Cost at $q^*$ = Variable Profit at $q^*$

First-Order Condition: Method 1

# 1) Use the MR and MC curves to illustrate variable profit



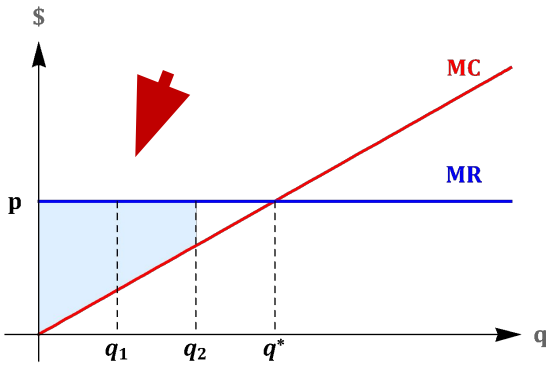



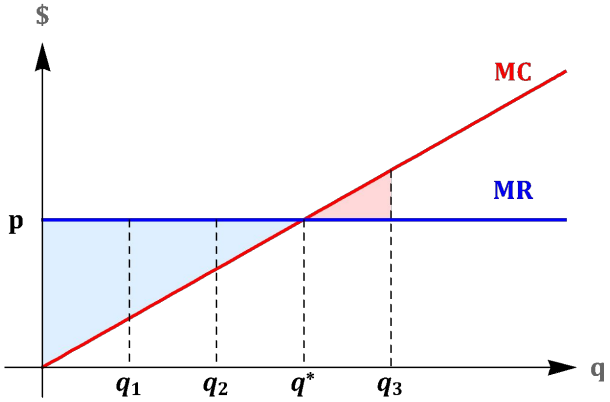

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First-Order Condition: Method 1

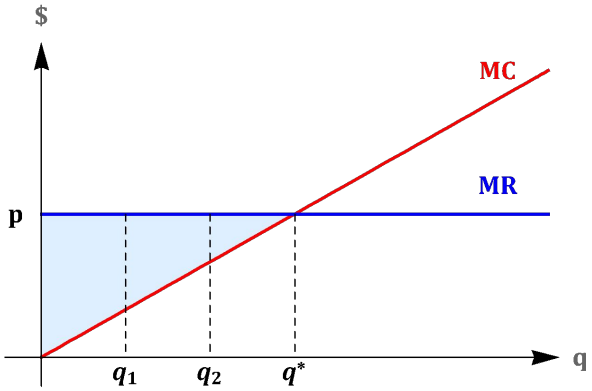
# 1) Use the MR and MC curves to illustrate variable profit




$q_2$ is too low



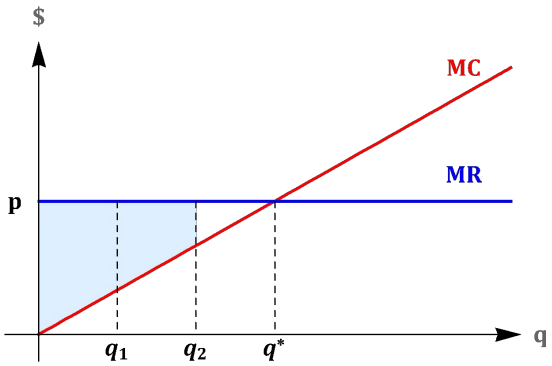

$q_3$ is too high



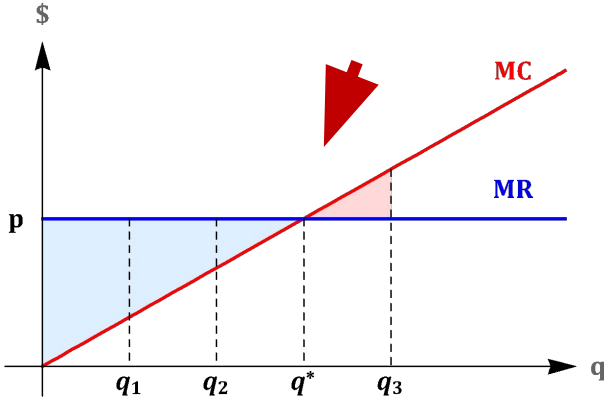

$q^*$ is optimal

First-Order Condition: Method 1

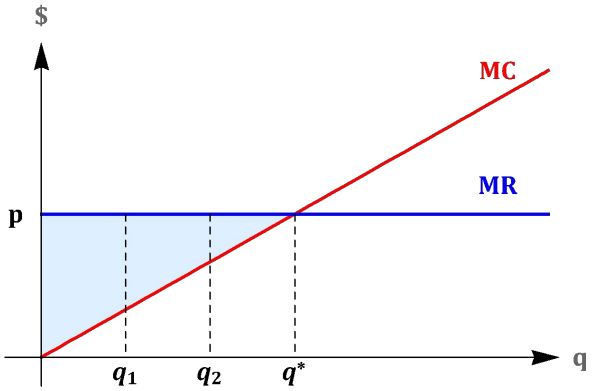
# 1) Use the MR and MC curves to illustrate variable profit




$q_2$ is too low

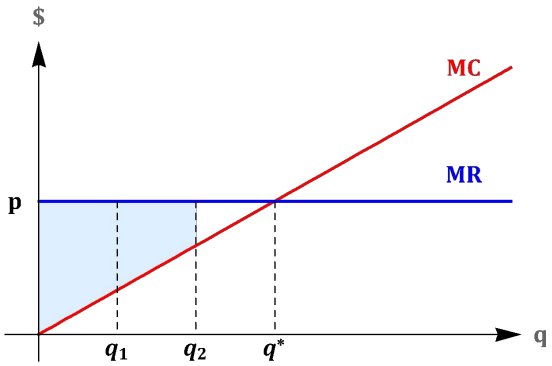



$q_3$ is too high

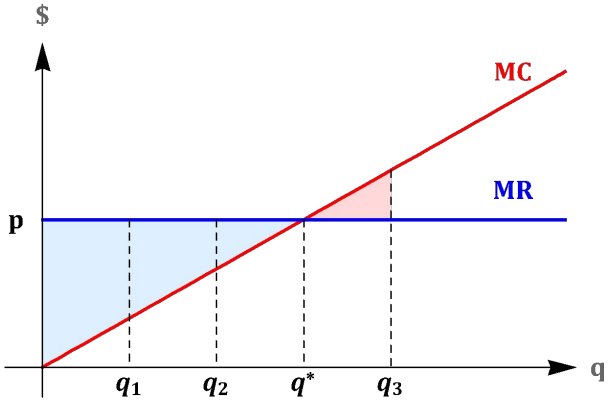



$q^*$ is optimal

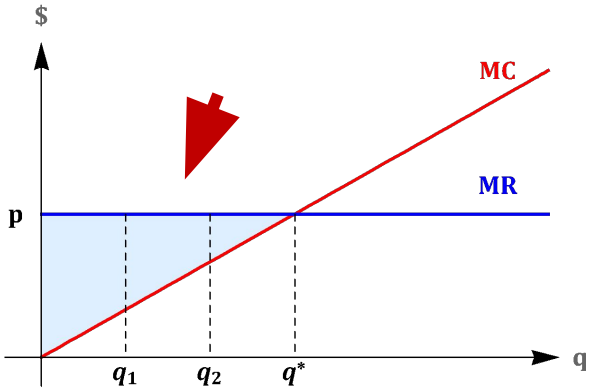
# 1) Use the MR and MC curves to illustrate variable profit




$q_2$ is too low




$q_3$ is too high




$q^*$ is optimal

# Graphic

https://www.econgraphs.org/textbook/firm_theory/profit_max/competitive_firm

# 2) Use the average cost (AC) curve to illustrate total profit

$$AC = \frac{C}{q} \rightarrow C = qAC$$

$$Profit(q) = Revenue - Cost$$

$$Profit(q) = pq - qAC$$

$$Profit(q) = (p - AC)q$$

profit per unit (average profit)

sales volume

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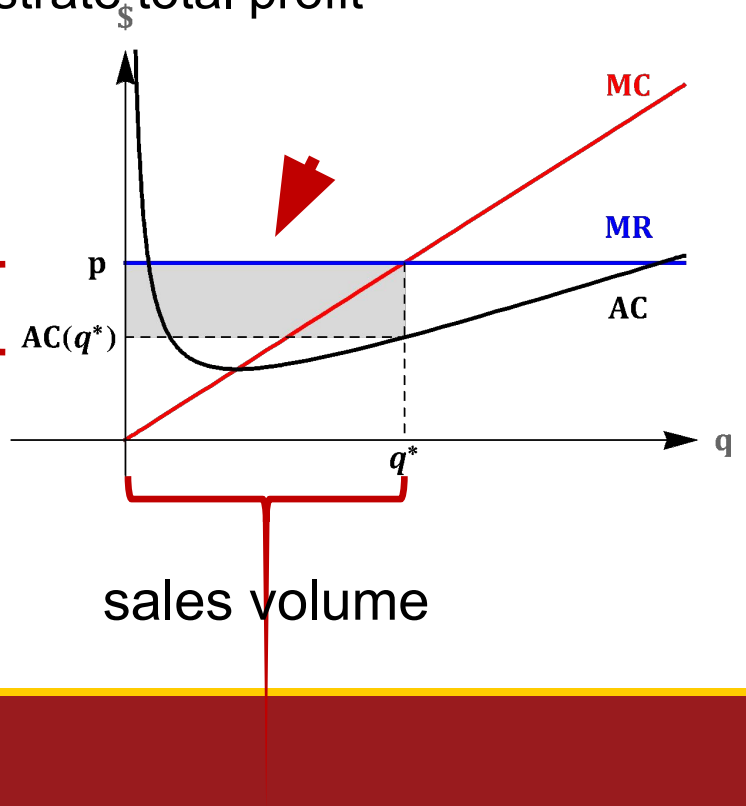
profit per unit (average profit)

sales volume

# 2) Use the average cost (AC) curve to illustrate total profit

$$Profit(q) = (p - AC)q$$

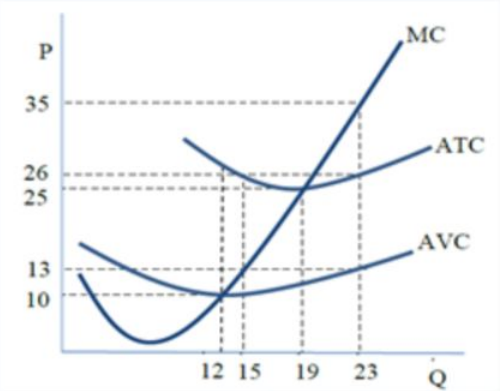
profit per unit
(average profit)



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# What quantity will maximize profit for the firm if p=13




0

12

15

19

23

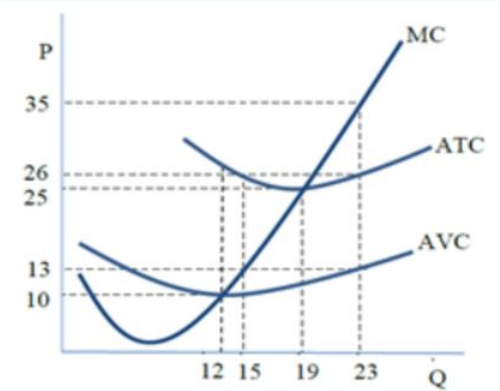
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# What quantity will maximize profit for the firm if p=35




0

12

15

19

23

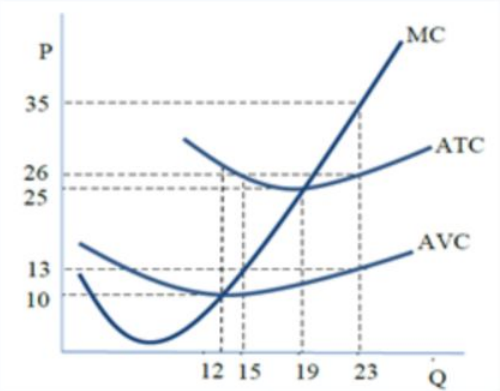
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# What is the profit for the firm if p=35?




805

207

23

230

275

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# Marginal cost, Average Cost and Price

- In a competitive market, firm will always choose q such that P=MC
- At the optimal quantity:
  - P < ATC , Profit < 0
  - P=ATC, Profit =0
  - P>ATC, Profit>0

Microeconomics
for Business

Chapter 3 – Production Choices

Maximizing Profit: Shutdown rule (short run)

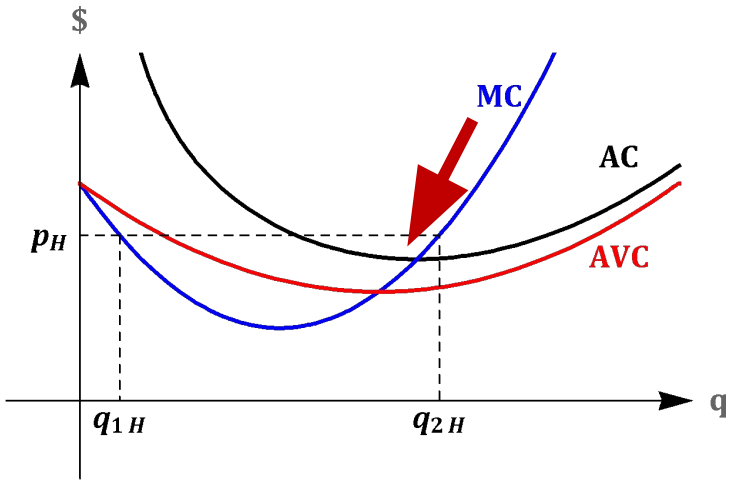
## Second-Order Condition: Two Checks

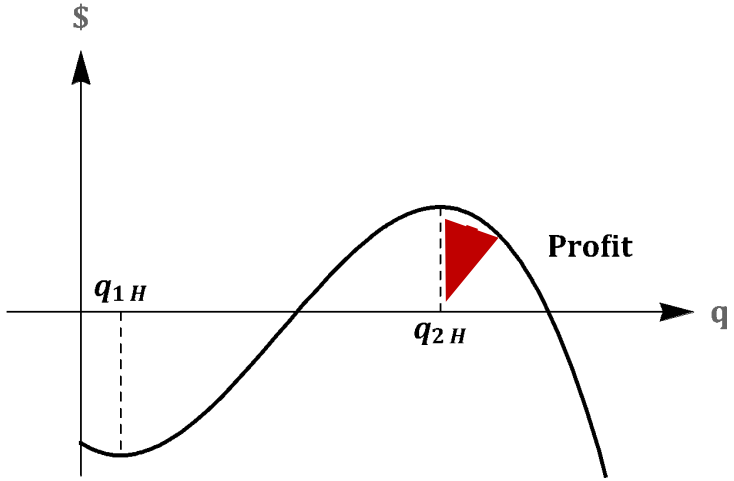
1) For a positive $q$ to a be local maximum, we need $p \geq$ minimum $AVC$ and we need $MC(q)$ to be increasing at $q$ where $p = MC(q)$

2) For shutdown to be optimal (a corner solution $q = 0$), $p <$ minimum $AVC$

Example 1: High price $p_H$: $p >$ minimum $AC$

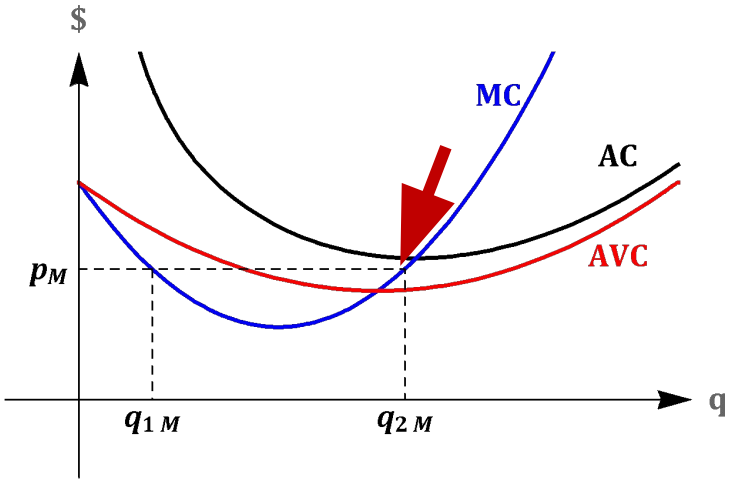
FOC $p_H = MC(q)$

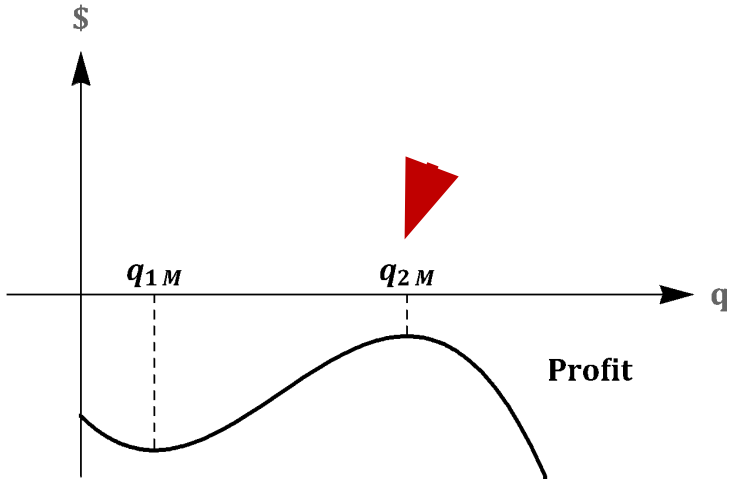


Example 2: Medium price $p_M$: minimum $AC > p >$ minimum $AVC$

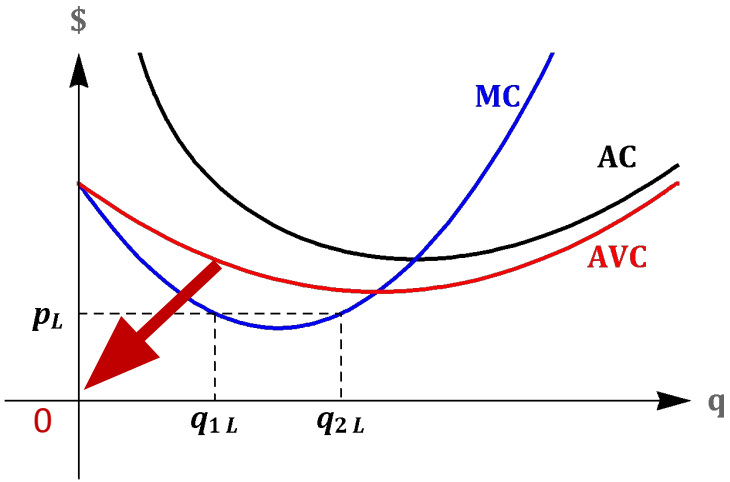
FOC $p_M = MC(q)$

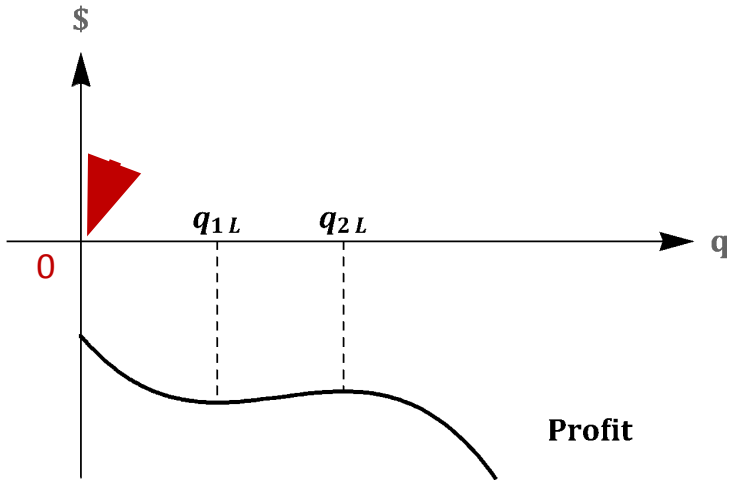


## Example 3: Low price $p_L$: $p <$ minimum $AVC$

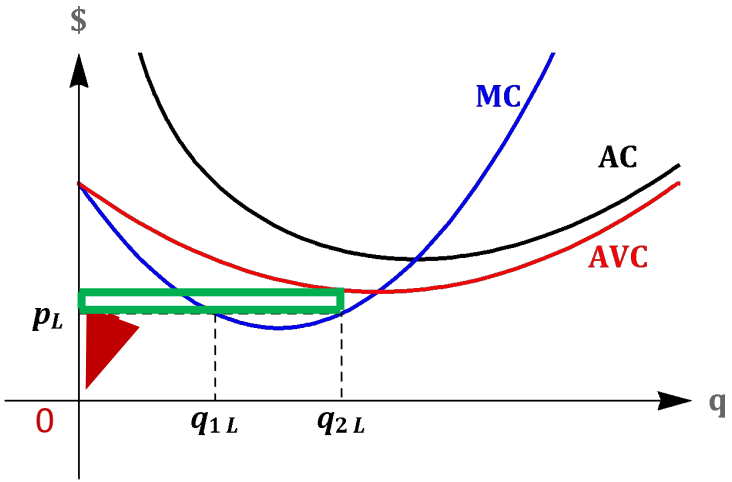
FOC $p_L = MC(q)$

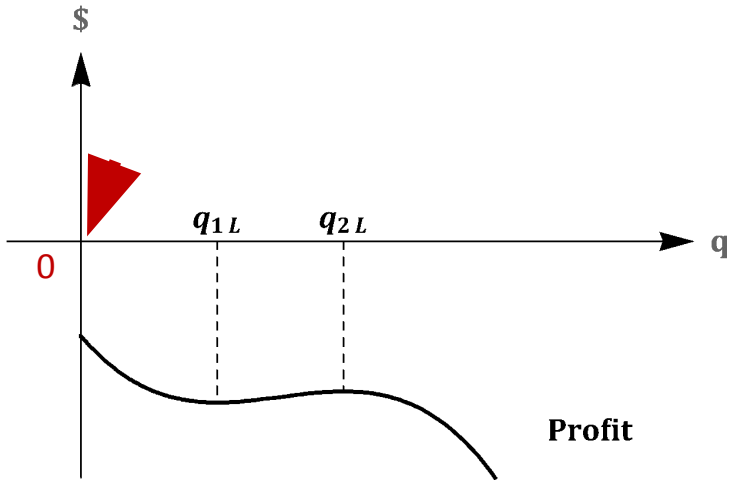


# Example 3: Low price $p_L$: $p <$ minimum $AVC$

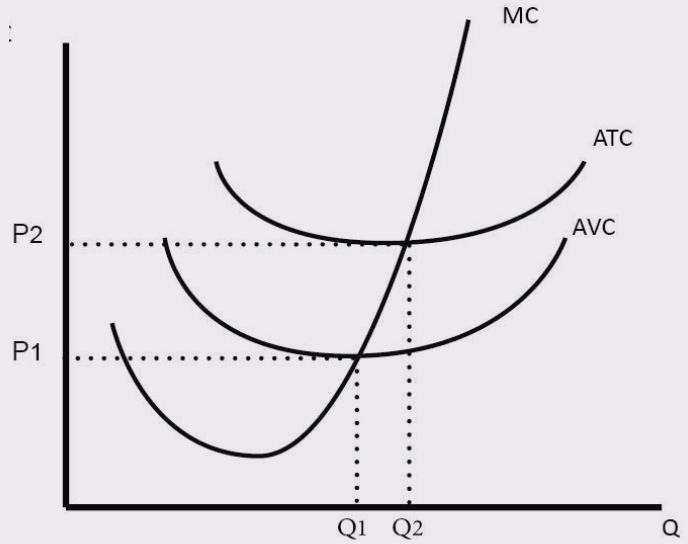
FOC $p_L = MC(q)$



## Second-Order Condition

If Price is below AVC, then the firm shuts down




Note: The average cost functions reach their minimum at the intersection with marginal cost.

# Cost Curves

Firms seek to answer 3 questions for any given market price:

- Below which price should the firm shutdown?
- At P2, how much should the firm produce?
- At P2, how much profit will the firm earn?

# Microeconomics for Business

## Chapter 3 – Production Choices

## The Firm's Supply Curve to Market Supply Curve and Market Producer Surplus

What is a supply curve?

What are supply shifters?

To maximize profits, the firm will produce quantity $q^*$ as follows:

Shutdown-Rule: the firm will produce $q^* = 0$ if the price $\mathrm{p}$ is below the minimum average variable cost.

If $\mathrm{p}$ is above the minimum AVC, then the firm produces $q^*$ such that

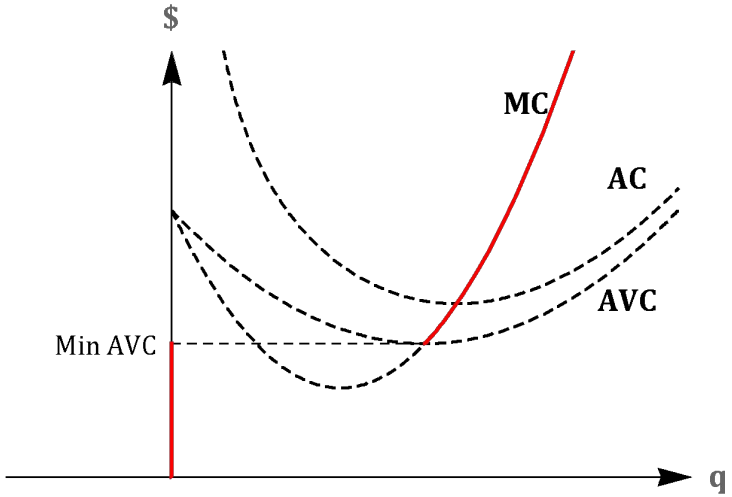
$$p = MC(q^*)$$

and $MC(q^*)$ is increasing at $q^*$

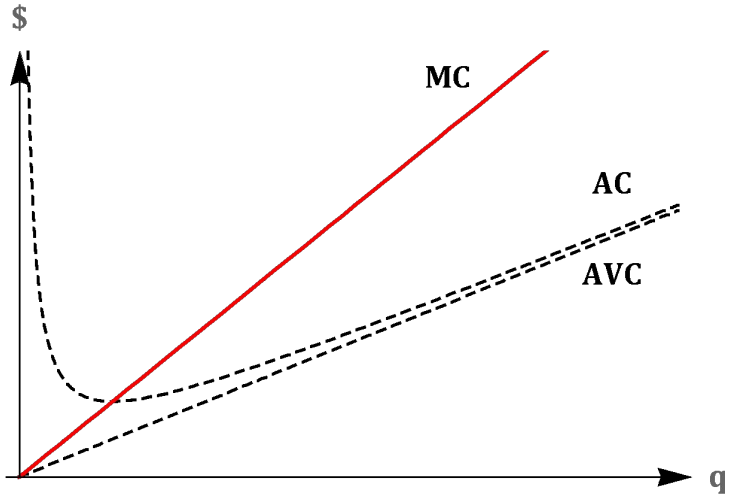
Supply Curve:

Graph mapping the market price $p$ into the quantity $q$ produced by the firm

# Example 1: Firm shuts down at low prices

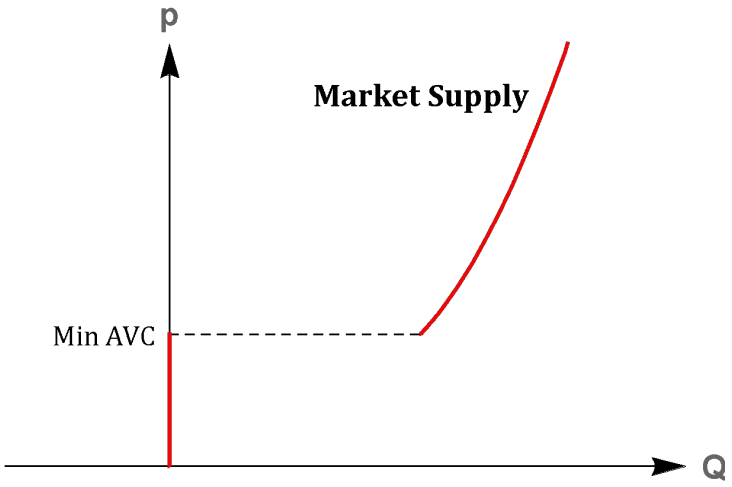


# Example 2: Firm never shutdowns

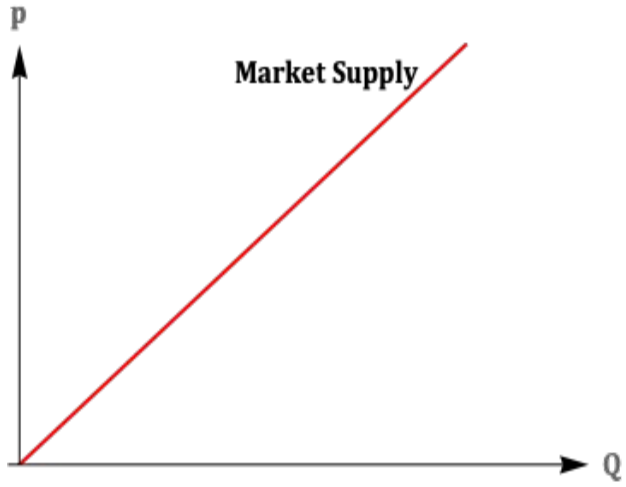


# Examples:

Shutdown




No shutdown



## Market Supply: Numerical Example

Market Supply is the horizontal sum of the individual supply schedules of all firms in the market

Example:

Each firm has a supply curve $q^S(p) = 2p$

There are 100 identical firms in the market

Market Supply is the horizontal sum of the individual supply schedules of all firms in the market

Example:

Each firm has a supply curve $q^S(p) = 2p$

There are 100 identical firms in the market

Market Supply: $Q^S(p) = 100 \times q^S(p) = 100 \times 2p = 200p$

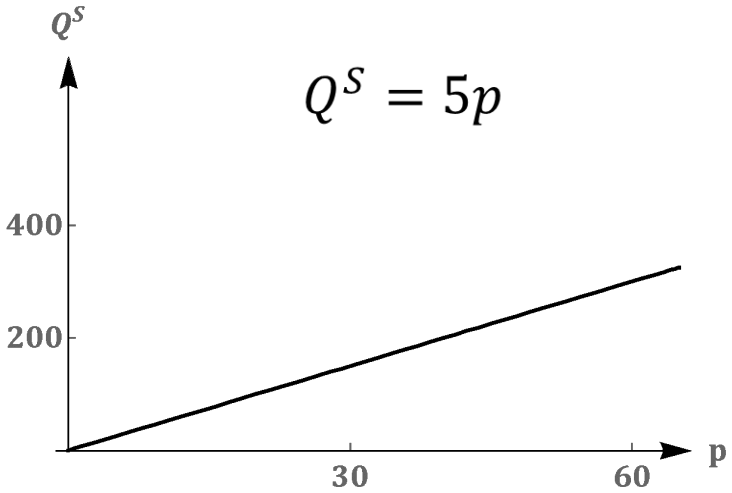
# Supply and Inverse Supply:

$$Q^S = 5p \qquad \text{Supply}$$

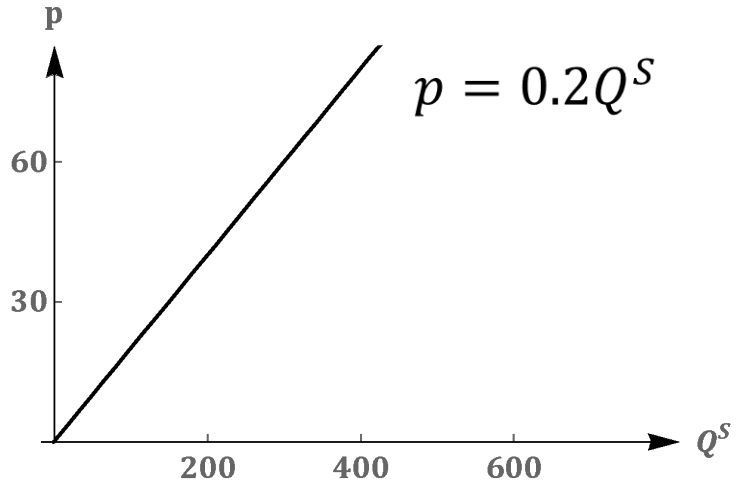
$$5p = Q^S \qquad \text{Solve Supply for p}$$

$$p = 0.2Q^S \qquad \text{Inverse supply}$$

## Market Supply versus Inverse Supply

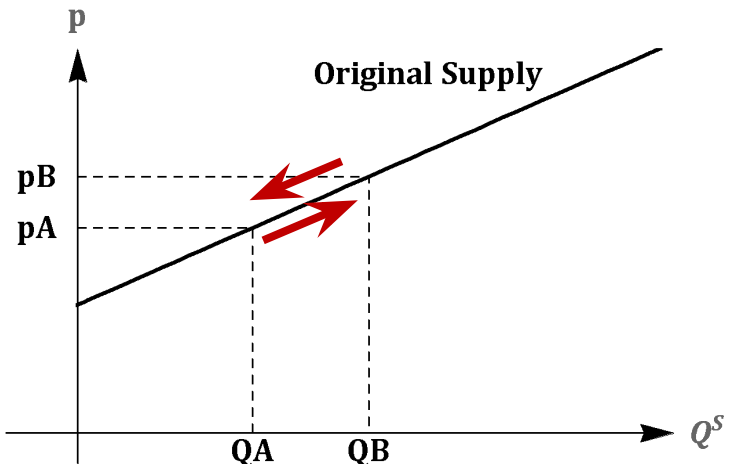



Supply




Inverse Supply

# Movement along the market supply curve: a price change

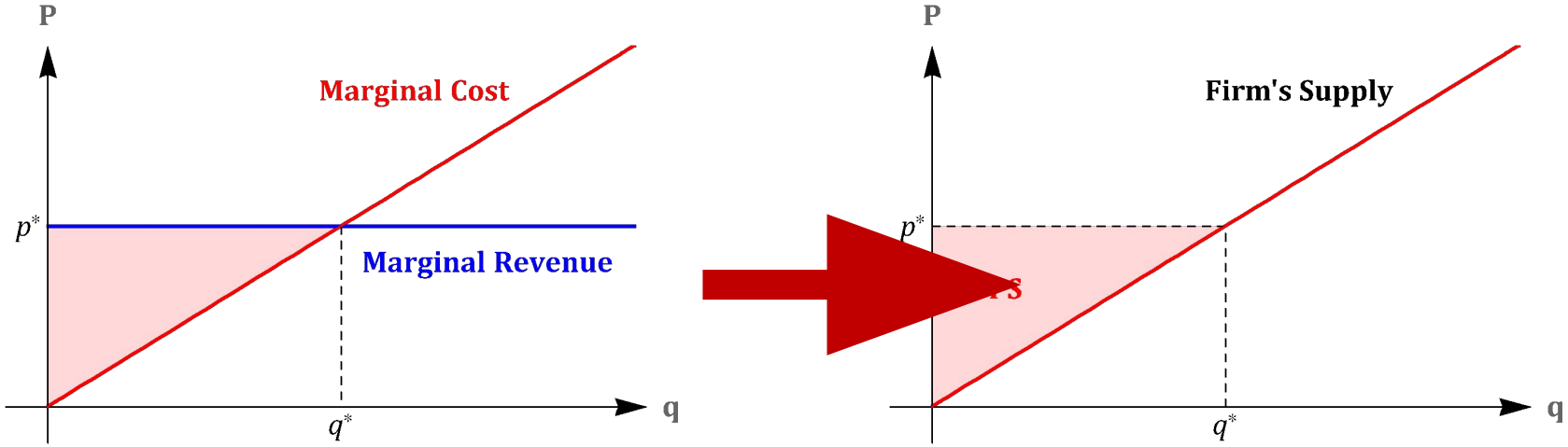


Producer surplus for a firm producing output $q$ is variable profit at this output.

Producer surplus is a measure of the value created by the firm when it produces $q$, not taking into account fixed costs.

# What is producer surplus and how is it calculated?

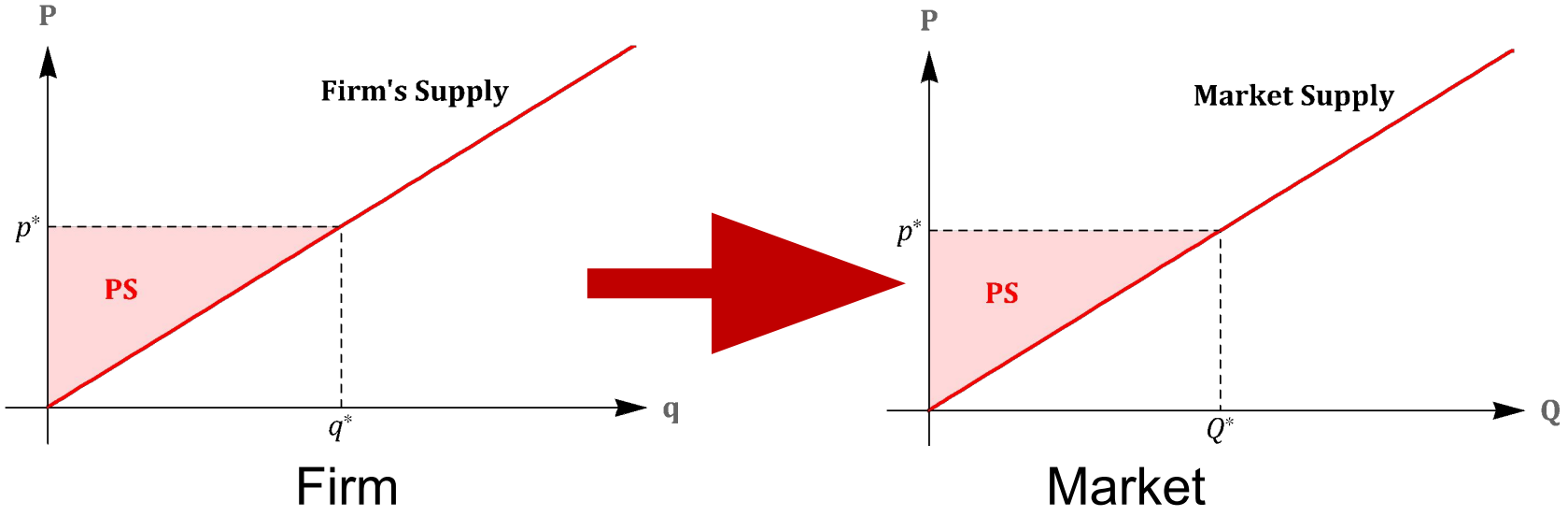
## Producer Surplus




Variable Profit

Producer Surplus

## Producer Surplus



# Problem

There are 100 identical firms with supply curve q=0.25p

P=10, what is total producer surplus?